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Status	Finished
Started	Thursday, 9 October 2025, 12:45 PM
Completed	Thursday, 9 October 2025, 1:01 PM
Duration	15 mins 50 secs
Marks	1.25/3.00
Grade	1.25 out of 3.00 (41.69%)

Information

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This page contains all the problems for this test. The very last problem asks you to contact the person in charge of the exam and tell him or her the 4-digit key given in the problem text. In return you will be given a 5-digit signing code which you must give as the answer to the problem.

This problem does not count towards the final score, but **tests missing this code will not count towards the final grade.**

The following rules apply:

- Total time allowed: 30 minutes. The test will automatically close if time runs out.
- UiA's usual rules in regards to cheating on exams apply.

Question **1**

Incorrect

Mark 0.00 out of 1.00

Consider the following regular expression.

$ex = Y^{**}[a-j]^{**}$

Which of the following regular expressions accepts at least the same words as ex ?

- ☒ $Y?[a-j]?$
- ☒ $Y^{*}[a-j]^{*}$
- ☐ $Y+[a-j]^{*}$
- ☐ $Y^{*}[a-j]^{+}$
- ☐ $Y+[a-j]?$
- ☒ $Y^{*}[a-j]?$
- ☐ $Y+[a-j]^{+}$
- ☒ $Y?[a-j]^{*}$
- ☐ $Y?[a-j]^{+}$
- ☐ None of the above

Question 2

Partially correct

Mark 0.25 out of 1.00

Consider the following specification.

 $\{c \leq 8\} \ c, z := c', z' \mid z > 14$

Which of the following implementations are correct with respect to the specification?

- ☐ $\{c \leq 8\} \ c := 16; z := c+3 \mid z > 14$
- ☒ $\{8 < 10\} \ c := 16; z := c+3 \mid z > 14$
- ☐ $\{c \leq 8\} \ c := 16; z := c+3 \mid z > 17$
- ☐ $\{c < 8\} \ c := 16; z := c+3 \mid z > 14$
- ☐ $\{c < 8\} \ c := 16; z := c+3 \mid 17 > 14$
- ☒ $\{8 < 10\} \ c := 16; z := c+3 \mid 17 > 14$
- ☐ $\{c \leq 8\} \ c := 16; z := c+3 \mid 17 > 14$
- ☐ $\{c < 8\} \ c := 16; z := c+3 \mid z > 17$
- ☒ $\{8 < 10\} \ c := 16; z := c+3 \mid z > 17$
- ☐ None of the above

Question 3

Correct

Mark 1.00 out of 1.00

Given the set $\{1, 2, 3, 4, 5\}$, and relations $\{[1, 1], [4, 3], [4, 1], [5, 2], [5, 5], [5, 1]\}$.

- Fill in the relations matrix below such that all given relations are marked by a "1" in the correct position.
- Then extend the set of relations by adding mandatory "1"s so that the final set of relations is a partial order.
- All other entries in the matrix must be marked "0".

-	1	2	3	4	5
1	1	0	0	0	0
2	0	1	0	0	0
3	0	0	1	0	0
4	1	0	1	1	0
5	1	1	0	0	1

Your last answer was interpreted as follows:

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 \end{bmatrix}$$

Question **4**

Correct

Mark 0.00 out of 0.00

Signing code

Before closing the test you must answer this problem with a signing code given to you by the person in charge of the test.

Tests missing this signing code will be ignored and will not count towards the final score.

Key: 526

Signing code:

Your last answer was interpreted as follows:

50151

[◀ Test 1 \(topics 1-3: Introduction, Concepts, Induction, Recursion, Grammars\)](#)

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